Alex Shengzhi Li

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Experience

Reka AI *Member of technical staff, Aug 2024 - Current*

Post-training modeling and evaluation to improve multi-modal instruction following and multi-lingual

TIFIN *Staff LLM Data Scientist, April 2024 - Present*
Senior NLP Scientist, Aug 2023 - April 2024

Owned the in-house architecture design of an agent-native framework for dynamic tool use, wrote from scratch, and launched in a pilot with JPMorgan and 50K users with the consumer app Magnifi.
Owned the finetuning of large-language models and their quality and consistency metrics. Applied SFT for tool use and SFT/DPO hybrid for response generation. Introduced constrained decoding for guaranteed function and parameter recall for production-grade quality from a Mistral-7B model.
Key contributor to document parsing and linearization. For RAG use cases, finetuned embedding model and re-ranker model which improved the quality metrics from 60-80%.
Developed wealth-specific benchmarks and scoped scaling law and training plan for in-house foundation model (continued pre-training of embedding models).
Re-designed the LLM technical screening guidelines and conducted more than 20 interviews. Organized inter-team knowledge share-out and journal club.

Cala Health *Data Scientist, Oct 2019 – Aug 2023*

Designed and implemented a therapy recommendation system using embedding search and contrastive pretraining techniques, contributing to a key milestone towards $77M funding and obtaining 3 patents.
Led data quality improvement using custom Wavelet-Spectral Unsupervised Clustering algorithm on motion data to filter out unwanted segments, increasing actionable sessions by 16%.
Owned a clinical study of motion monitoring for Medicare approval, developing an activity recognition model that outperformed baseline models by 12% with large-scale self-supervised learning on UK-BioBank 100K patient data

Zeesan Biotech *Data Scientist, June 2018 - Oct 2019*

Decreased end-user re-verification rates by 35% through a feature-engineering and ensemble-tree model for time series classification task, reducing support engineering efforts.
Introduced and implemented SHAP, a feature importance calculation tool for explainable biomarker discovery.

Education

WestCliff University *May 2021 - May 2022*

Master of Science in Computer Science, Summa Cum Laude

Stanford University *Jan 2021 - Dec 2021*

Graduate Certificate in Artificial Intelligence

University of California, Berkeley *Aug 2013 - May 2018*

Bachelor of Arts in Applied Mathematics
Bachelor of Arts in Physics

Skills

Tools: Python, C++, PyTorch, TensorFlow, PyTorch Lightning, AWS (SageMaker), GCP
Data Science Tools: Numpy, Spark, SQL, RAPIDS

Containerization and Deployment: Docker, Flask, VLLM, TRT-LLM, SageMaker, Gitlab CI/CD
Machine Learning Ecosystem: HuggingFace, DeepSpeed, NeMo
Large Language Models: DPO, SFT (Functional Calling, Text2SQL), Advanced RAG
LLM Specializations: Agentic Workflows, Multi-modal LLM, Constrained Decoding

Publications

Li, S., & Tajbakhsh, N. (2023). ScigraphQA: A large-scale synthetic multi-turn question-answering dataset for scientific graphs. arXiv:2308.03349. GitHub Huggingface Downloaded 6K+ April 2024
Li, S., Lin, R., & Pei, S. (2024). Multi-modal preference alignment remedies regression of visual instruction tuning on language model. arXiv:2402.10884 HuggingFace. Accepted into ACL 2024 Main Conference (top 10% score).
Li, S., Lin, R., Kampa, K., Bohang, L., Pei, S., & Han, B. (2024). You Comprehended the Whole Paper: Academic Reviews Enhance LLM Long-Context Capabilities. Under review at EMNLP 2024.

Reviewing Experience

NeuralIPS 2024
IEEE Transactions on Intelligent Vehicles (Impact Factor: 8.2)
Neural Networks (Impact factor 7.8)
Computers and Electrical Engineering (Impact Factor: 4.3)

Patents

Kent, A., Schulte, G., Li, S., et al. (2022). Parameter variations in neural stimulation. WO2023015159A3.
Schulte, G., Li, S., Kent, A., (2022) Personalized therapy neuromodulation systems. WO2023283568A1.
Balbaky, S., ... Li, S., Wells, K. (2022). Wearable neurostimulation System. AU2022324555A1.

Conference Talks

Democratize 24, San Francisco. "The long research winter from 2012 to 2022.". Link
The International Tremor Congress 2023, New York. "Optimization of Non-invasive Transcutaneous Afferent Patterned Stimulation (TAPS) Therapy in Essential Tremor." Link

Organizational Experience

Research to the People, Stanford 2023. Invited to co-organize Rare Disease AI Hackathon at Stanford School of Medicine Link
Cal Hacks, UC Berkeley 2018. Mentor for the world's largest collegiate hackathon